# Extrauterine Pregnancy 09. Cesarean Scar Pregnancy: Early PAS Thinking

Companion reading handout for simultaneous listening.
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## Transcript

Extrauterine Pregnancy audio course. Episode 9. Cesarean Scar Pregnancy: Early PAS Thinking.

Cesarean scar ectopic pregnancy is an early pregnancy implanted in or in close contact with the niche of a previous cesarean delivery scar. It is inside the uterus in a geographic sense, but it is not normally sited in the endometrial cavity. The clinical danger comes from that distinction. The trophoblast is not implanted into ordinary decidualized endometrium over healthy myometrium; it is implanted into scar tissue where decidualization may be defective and the residual myometrium may be thin or absent.

That implantation bed explains why cesarean scar pregnancy belongs next to placenta accreta spectrum in the resident's mind. ISUOG states that cesarean scar pregnancy can lead to severe hemorrhage, uterine rupture, and development of PAS disorders. SMFM describes expectantly managed cases as having high rates of severe morbidity, including hemorrhage, PAS, and uterine rupture. Histopathologic work cited by ISUOG and SMFM supports the same conceptual link: scar pregnancy and early placenta accreta can share common features.

This is the first repetition the listener should keep. Scar pregnancy is not just "an ectopic that happens to be low." It is early abnormal placentation in a damaged implantation bed. The scar niche may lack normal decidua, may have deficient myometrium, and may sit near large vessels and bladder. Once that is understood, the later management choices sound less arbitrary. The problem is not only whether the embryo is alive. The problem is where the placenta is anchoring.

The diagnosis begins with a low anterior implantation in a patient with a prior cesarean delivery. The classic ultrasound pattern is an empty uterine cavity, an empty endocervical canal, and a gestational sac, placenta, or trophoblast embedded at the hysterotomy scar. SMFM describes a triangular sac at or before 8 weeks, or a rounded or oval sac after 8 weeks, filling the scar niche; a thin, 1 to 3 millimeter, or absent myometrial layer between the sac and bladder; and prominent vascularity at the scar on color Doppler.

The numbers belong to the image. One to 3 millimeters of myometrium between the sac and bladder is not a decorative measurement. It tells you how little contractile uterine muscle separates trophoblast from the anterior serosa and bladder. Before 8 weeks, the sac may still show the triangular niche relationship clearly. After 8 weeks, the sac may look rounder and more intracavitary. The earlier scan is therefore not just earlier in time; it is often more honest about the implantation site.

The scan should not stop after saying "intrauterine." In a patient with a prior cesarean, the relevant question is where the sac and trophoblast are located relative to the endometrial cavity, the lower uterine segment, the scar niche, the cervix, the bladder, and the anterior myometrium. Jordans and the Delphi group emphasize reporting the niche, residual myometrial thickness, adjacent myometrial thickness, gestational sac size, vascularity, trophoblast invasion into myometrium, relationship to serosa and bladder, and relation to uterine vessels. These are not decorative ultrasound measurements. They describe the surgical and hemorrhage problem.

Time changes the appearance. ISUOG stresses that prenatal diagnosis is easier before 9 weeks. As gestation advances, the upper pole of the sac may grow toward the fundus and appear more intracavitary, while the placenta remains anchored in the scar niche. That is the source of a dangerous false reassurance: the embryo seems to have moved into the uterine cavity, but the placental bed has not become normal. The anchor, not only the sac contour, defines the disease.

This before-9-weeks point should stay in memory because it changes practice. The scan in a patient with previous cesarean should actively look for scar implantation early, while the sac still reveals the niche relationship. Later, the fetus may be seen higher in the cavity, but the placenta can remain low, anterior, vascular, and dangerous. In scar pregnancy, the fetus can move visually; the placental anchor is the truth.

The main differentials are cervical pregnancy and miscarriage in progress. Cervical pregnancy is below the internal os. Cesarean scar pregnancy is usually at or above the internal os and anterior at the scar. A miscarriage passing through the cervix or temporarily lodged near the scar should move with gentle probe pressure and usually lacks peritrophoblastic blood flow. Farren emphasizes the sliding sign and Doppler vascularity: short-lived migration slides and is not vascularly anchored; true scar implantation is fixed and vascular.

Management follows from the tissue bed. SMFM recommends against expectant management of recognized cesarean scar ectopic pregnancy. This is a strong recommendation because continuation can lead to PAS, massive hemorrhage, uterine rupture, hysterectomy, severe morbidity, and maternal death. SMFM notes an exception for a definitively nonviable early scar pregnancy, where serial ultrasound, quantitative hCG, and symptom monitoring may be pursued, but even then resolution can take months and complications such as uterine arteriovenous malformation have been described.

The recommendation against expectant management should not be memorized as a blunt prohibition detached from biology. It is the consequence of a pregnancy that may turn into a PAS level placental problem. If the placenta remains implanted in a scar niche, continued gestation can increase vascularity, deepen invasion, and make later separation catastrophic. Expectant management of a recognized live scar pregnancy is therefore not like watching a low-hCG failing ampullary ectopic.

Treatment choice depends on gestational age, viability, vascularity, depth of implantation, residual myometrium, symptoms, patient goals, local expertise, and available hemorrhage-control tools. SMFM suggests operative resection, by transvaginal or laparoscopic approaches when possible, or ultrasound-guided uterine aspiration. It specifically advises avoiding sharp curettage alone. The reason is anatomical: sharp curettage scrapes the cavity, but the target tissue is in a vascular scar defect outside the normal endometrial plane. Scar tissue contracts poorly, deeply invading vessels may be opened, and the trophoblast may not be completely accessible from the cavity.

Farren describes why surgical blood loss can be so severe: large arcuate and helicine arteries, loss of myometrium in the affected area, and reduced contractility. Transcervical suction evacuation under continuous ultrasound guidance can be logical for partial scar pregnancies that extend into the cavity, especially when combined with hemostatic measures such as balloon tamponade, Shirodkar suture, uterotonics, uterine artery embolization, or surgical backup. Complete scar pregnancies or pregnancies growing toward the bladder or broad ligament may require transabdominal resection or more complex surgery.

Methotrexate also has to be understood through anatomy. SMFM recommends intragestational methotrexate, with or without other modalities, when medical treatment is chosen, but recommends against systemic methotrexate alone. Systemic methotrexate may suppress trophoblast, yet it does not directly solve the scar-bed hemorrhage problem. Farren reports systemic methotrexate effectiveness of only about 59 percent in an international registry and describes high complication risk. Local intragestational treatment delivers drug into the pregnancy itself, but resolution may still take weeks to months; SMFM cites a mean resolution time of 88 days in one local conservative series.

The contrast is important. Systemic methotrexate treats the whole patient and reaches the trophoblast indirectly. Intragestational methotrexate targets the pregnancy directly. Neither one stitches the scar, thickens the residual myometrium, or removes the vascular implantation bed. That is why scar-pregnancy treatment often combines methods: local treatment, image guidance, aspiration, resection, tamponade, embolization, or surgical backup. The plan is built around bleeding control as much as trophoblast suppression.

The wanted-pregnancy conversation is the hardest part of this lecture. A live embryo does not make the implantation safe. If a patient chooses continuation after counseling, the pregnancy should be managed as high-risk placental disease. SMFM recommends repeat cesarean delivery between 34 0/7 and 35 6/7 weeks, with betamethasone before the anticipated late-preterm delivery, in a level III or IV facility prepared for PAS level hemorrhage, massive transfusion, multidisciplinary surgery, and possible cesarean hysterectomy.

Counseling should use exact language. Tell the patient that the pregnancy is implanted in the cesarean scar niche rather than normally in the upper endometrial cavity. Explain that the placenta may grow into scar and myometrium in a way that prevents safe separation. Name the concrete risks: severe bleeding, transfusion, uterine rupture, hysterectomy, preterm delivery, loss of fertility, and maternal death. Also name the uncertainty: treatment options vary because high-quality comparative evidence is limited, and management often depends on anatomy and local expertise.

Future pregnancy counseling is part of the diagnosis. SMFM recommends counseling about recurrence and effective contraception, including long-acting reversible contraception and permanent contraception when appropriate. ISUOG reports recurrence after prior scar pregnancy in published literature and recommends early ultrasound in subsequent pregnancy, ideally before 8 weeks, to confirm normal location and exclude recurrent scar implantation or PAS.

The resident behavior is straightforward: in every early pregnancy after cesarean delivery, do not be satisfied by the word "intrauterine." Identify the implantation site, the placental anchor, the residual myometrium, the relationship to the bladder and cervix, and the Doppler vascularity. The disease is not simply a pregnancy in the uterus. It is abnormal implantation in a scar.